

LM05 Analyzing Statements of Cash Flows II

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1. Introduction

This learning module covers the tools and techniques for analyzing the statement of cash flows such as:

- Evaluating the sources and uses of cash
- Common size analysis
- Calculation of free cash flow measures
- Cash flow ratios

2. Evaluating Sources and Uses of Cash

Evaluation of the cash flow statement should involve the following:

- Evaluate where the major sources and uses of cash flow are between operating, investing, and financing activities. Major sources of cash for a company can vary with its stage of growth. For example, for a mature company it is expected that operating activities are the primary source of cash flows. However, analysts must analyze whether operating cash flows are positive and cover capital expenditures for all companies.
- Evaluate the primary determinants of operating cash flow. Analysts should compare operating cash flow with net income. If a company has large net income but poor operating cash flow, it may be a sign of poor earnings quality. In addition, analysts need to look at consistency of operating cash flows.
- Evaluate the primary determinants of investing cash flow. This is useful for letting the analyst know how much is being invested for the future in property, plant, and equipment and how much is put aside in liquid investments.
- Evaluate the primary determinants of financing cash flow. This helps understand if the company is raising capital or repaying capital and which capital sources are being used.

3. Ratios and Common-Size Analysis

In common-size analysis of a company's cash flow statement, there are two alternative approaches. In the first approach, we express each line item of cash inflow (outflow) as a percentage of total inflows (outflows). An example of a common-size cash flow statement using this approach is shown below for a fictitious company - K2 Corp.

Inflows	Actual	% of Total Inflow
Net cash provided by operating activities	3,200	80 %
Sale of Equipment	800	20%
Total	4,000	100%
Outflows	Actual	% of Total Outflow

Purchase of equipment	1,500	36.58%
Retirement of long-term debt	500	12.19%
Retirement of common stock	325	7.9%
Dividend payments	1,775	43.29%
Total	4,100	100%
Net increase (decrease) in cash	(100)	

In the second approach, we express each line item as a percentage of revenue. An example of such a statement is shown below for K2 Corp. In this example, we have assumed total revenue is 10,000.

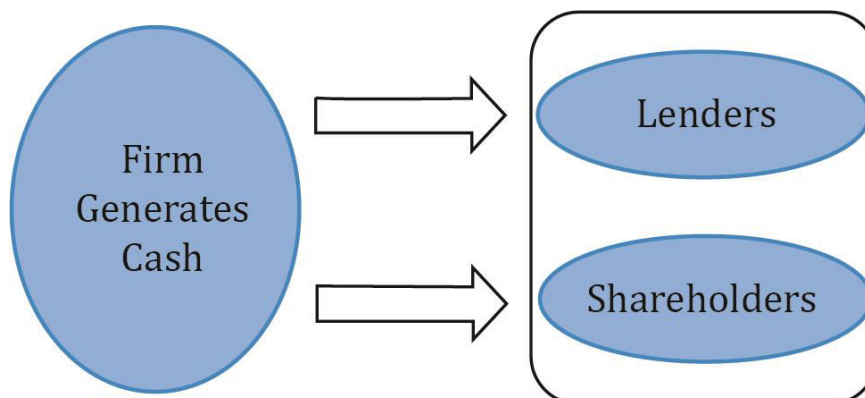
Cash flow	Actual	% of Total Revenue
Cash flow from operating activities		
Net Income	4,000	40%
Depreciation expense	500	5%
Increase in accounts receivable	(500)	(5%)
Increase in inventory	(1,000)	(10%)
Decrease in prepaid expenses	1,000	10%
Increases in accounts payable	500	5%
Increases in accrued liabilities	500	5%
Net cash provided by operating activities	5,000	50%
Cash flow from investing activities		
Cash received from sale of equipment	2,000	20%
Cash paid for purchase of equipment	(5,000)	(50%)
Net cash used for investing activities	(3,000)	(30%)
Cash flow from financing activities		
Sale of bonds	1,000	10%
Cash dividends	(2,000)	(20%)
Net cash used for financing activities	(1,000)	(10%)
Net increase in cash	1,000	10%

The common-size cash flow statement makes it easier to see trends in cash flow rather than just looking at the total amount. The second approach is useful for the analyst in forecasting future cash flows.

4. Free Cash Flow Measures

Free cash flow to firm (FCFF) is the cash flow available to all the suppliers of capital to a company after all operating expenses have been paid and necessary investments in working

capital and fixed capital have been made. The suppliers of capital include both lenders (debt) and equity shareholders (equity). This is illustrated in the figure below:



The formula for computing FCFF from net income is:

$$\text{FCFF} = \text{NI} + \text{NCC} + \text{Int} (1 - \text{Tax rate}) - \text{FCInv} - \text{WCInv}$$

where:

NI = Net Income

NCC = non-cash charges

Int = Interest expense

FCInv = Fixed capital investment/expenditures

WCInv = working capital expenditures

FCFF can also be computed from CFO as:

$$\text{FCFF} = \text{CFO} + \text{Int}(1 - \text{Tax rate}) - \text{FCInv}$$

While FCFF indicates how much cash is available to all suppliers of capital, **free cash flow to equity** (FCFE) is the cash flow available to the company's stockholders after all operating expenses and borrowing costs (principal and interest) have been paid and necessary investments in working capital and fixed capital have been made. The formula for computing FCFE is as follows:

$$\text{FCFE} = \text{CFO} - \text{FCInv} + \text{Net borrowing}$$

5. Cash Flow Statement Analysis: Cash Flow Ratios

There are several ratios useful for the analysis of the cash flow statement. These ratios generally fall into cash flow performance (profitability) ratios and cash flow coverage (solvency) ratios. The calculation and interpretation of these ratios are summarized in the tables below.

Performance Ratios	Calculation	What It Measures
Cash flow to revenue	$\text{CFO} \div \text{Net revenue}$	Operating cash generated per dollar of revenue.

Cash return on assets	$\text{CFO} \div \text{Average total assets}$	Operating cash generated per dollar of asset investment.
Cash return on equity	$\text{CFO} \div \text{Average shareholders' equity}$	Operating cash generated per dollar of owner investment.
Cash to income	$\text{CFO} \div \text{Operating income}$	Cash generating ability of operations.
Cash flow per share	$(\text{CFO} - \text{Preferred dividends}) \div \text{Number of common shares outstanding}$	Operating cash flow on a per-share basis.

Coverage Ratios	Calculation	What It Measures
Debt coverage	$\text{CFO} \div \text{Total debt}$	Financial risk and financial leverage.
Interest coverage	$(\text{CFO} + \text{Interest paid} + \text{Taxes paid}) \div \text{Interest paid}$	Ability to meet interest obligations.
Reinvestment	$\text{CFO} \div \text{Cash paid for long-term assets}$	Ability to acquire assets with operating cash flows.
Debt payment	$\text{CFO} \div \text{Cash paid for long-term debt repayment}$	Ability to pay debts with operating cash flows.
Dividend payment	$\text{CFO} \div \text{Dividends paid}$	Ability to pay dividends with operating cash flows.
Investing and financing	$\text{CFO} \div \text{Cash outflows for investing and financing activities}$	Ability to acquire assets, pay debts, and make distributions to owners.

Summary

LO: Analyze and interpret both reported and common-size cash flow statements.

Operating cash flow

- A healthy firm should generate positive cash flows from operating activities. (This is not applicable for startups).
- Positive operating cash flow generated by liquidating non-cash working capital items (like liquidating inventory and receivables or increasing payables) is not sustainable.
- Earnings that are significantly greater than operating cash flows indicate that aggressive accounting policies are being followed.

Investing cash flow

- Cash outflows can result from investments in property, plant and equipment, or other assets.
- Increasing outflows is an indication of growth.
- Decreasing outflows may indicate a reduction of capital expenditure and reduction in growth.

Financing cash flow

- Tells us if the company is generating cash by issuing debt or equity.
- Also tells us if the company is using cash to repay debt, reacquire stock, or pay dividends.

Common-Size format

- There are two approaches:
 - In the first approach, we express each line item of cash inflow (outflow) as a percentage of total inflows (outflows).
 - In the second approach, we express each line item as a percentage of revenue.
- Common-size cash flow statement makes it easier to identify trends in cash flows.
- It also helps us in forecasting future cash flows.

LO: Calculate and interpret free cash flow to the firm, free cash flow to equity, and performance and coverage cash flow ratios.

$$\text{FCFF} = \text{NI} + \text{NCC} + \text{Int} (1 - \text{Tax rate}) - \text{FCInv} - \text{WCInv}$$

Or

$$\text{FCFF} = \text{CFO} + \text{Int} (1 - \text{Tax rate}) - \text{FCInv}$$

$$\text{FCFE} = \text{CFO} - \text{FCInv} + \text{Net borrowing}$$

Commonly used cash flow ratios are:

		Calculation	What It Measures
Performance Ratios	Cash flow to revenue	$\text{CFO} \div \text{Net revenue}$	Operating cash generated per dollar of revenue.
	Cash return on assets	$\text{CFO} \div \text{Average total assets}$	Operating cash generated per dollar of asset investment.
	Cash return on equity	$\text{CFO} \div \text{Average shareholders' equity}$	Operating cash generated per dollar of owner investment.
	Cash to income	$\text{CFO} \div \text{Operating income}$	Cash generating ability of operations.
	Cash flow per share	$(\text{CFO} - \text{Preferred dividends}) \div \text{Number of common shares outstanding}$	Operating cash flow on a per-share basis.
Coverage Ratios	Debt coverage	$\text{CFO} \div \text{Total debt}$	Financial risk and financial leverage.
	Interest coverage	$(\text{CFO} + \text{Interest paid} + \text{Taxes paid}) \div \text{Interest paid}$	Ability to meet interest obligations.
	Reinvestment	$\text{CFO} \div \text{Cash paid for long-term assets}$	Ability to acquire assets with operating cash flows.
	Debt payment	$\text{CFO} \div \text{Cash paid for long-term debt repayment}$	Ability to pay debts with operating cash flows.
	Dividend payment	$\text{CFO} \div \text{Dividends paid}$	Ability to pay dividends with operating cash flows.
	Investing and financing	$\text{CFO} \div \text{Cash outflows for investing and financing activities}$	Ability to acquire assets, pay debts, and make distributions to owners.